

Synthesis and Characterization of Poly (Glycerol Dodacanedioate) Methacrylate as a Photocrosslinkable Shape Memory Biomaterial

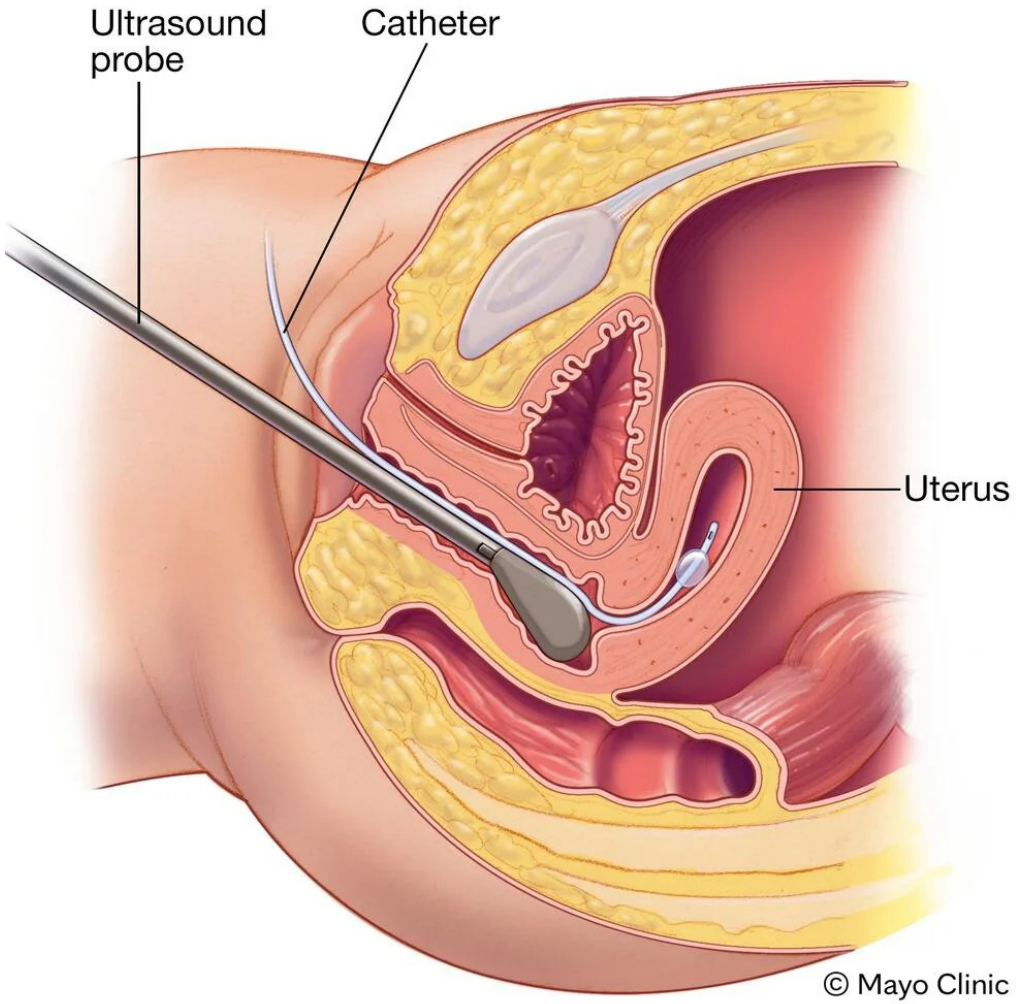
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BioEngineering for Translational Applications

Shape-memory polymers for minimally invasive surgery



Hysteroscopy in Uterus

Limitations of current SMPs

Don’t have transition temperature not near 37 °C (PGS)

Mechanical incompatibility with soft tissue (PCL, Polyurethane).

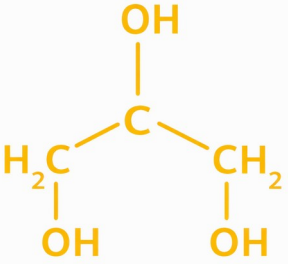
Limited biomedical applications.

Do we have a shape memory polymer that has transition temperature near body temperature & compatible with soft tissues?

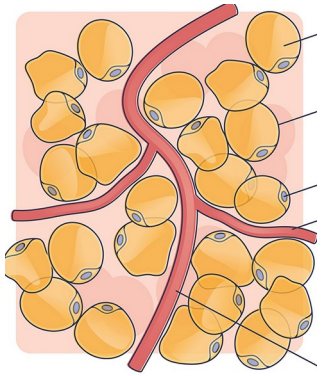
Poly Glycerol Dodacanedioate (PGD)

- T_g ~37 °C and nonlinear elasticity .
- Non-toxic monomers: Dodecanedioic acid and glycerol.

PGD is a thermoset polymer and need harsh curing conditions for further use.

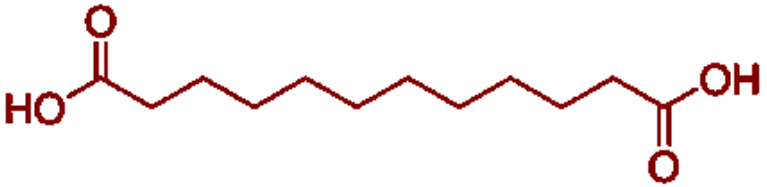


Glycerol

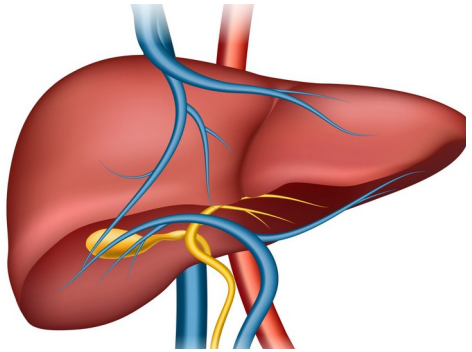


Glycerol is a metabolite in fat tissue

We need to introduce **photocurable chemistry** in PGD to use it in SLA & DLP



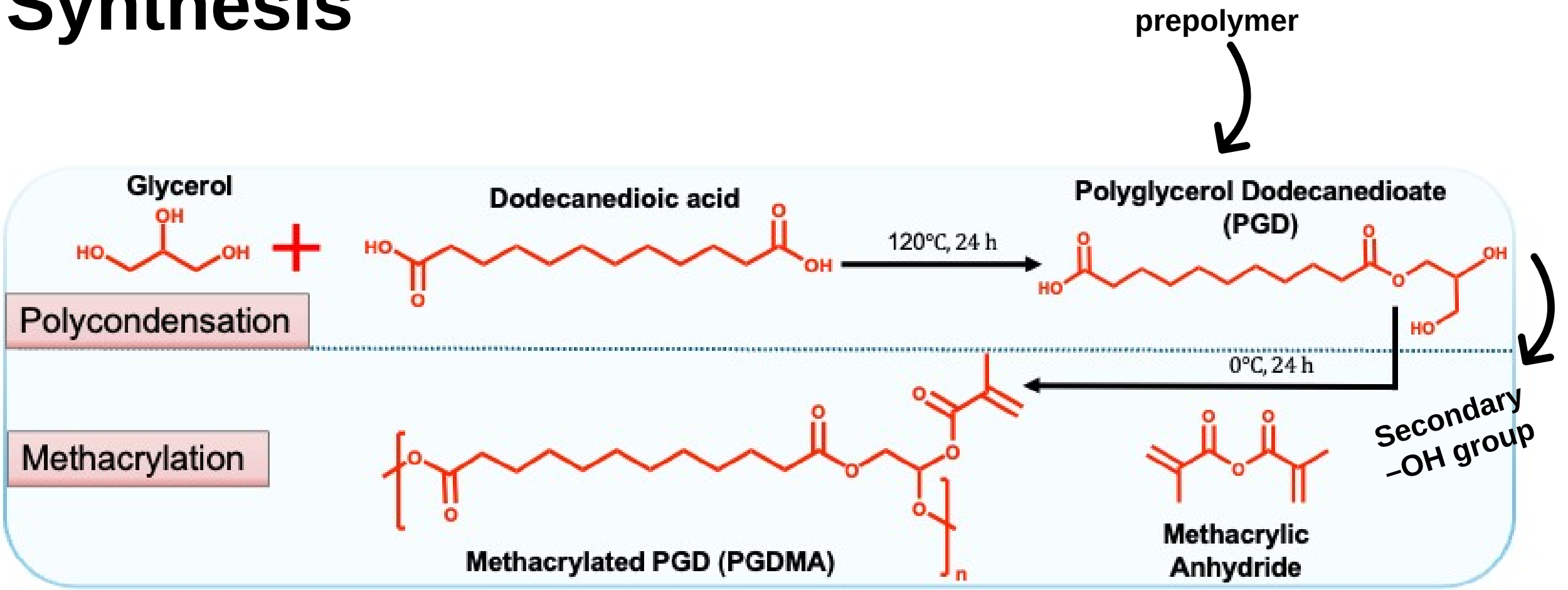
Dodecanedioic acid (DDA)



DDA is a metabolite in liver metabolic pathway

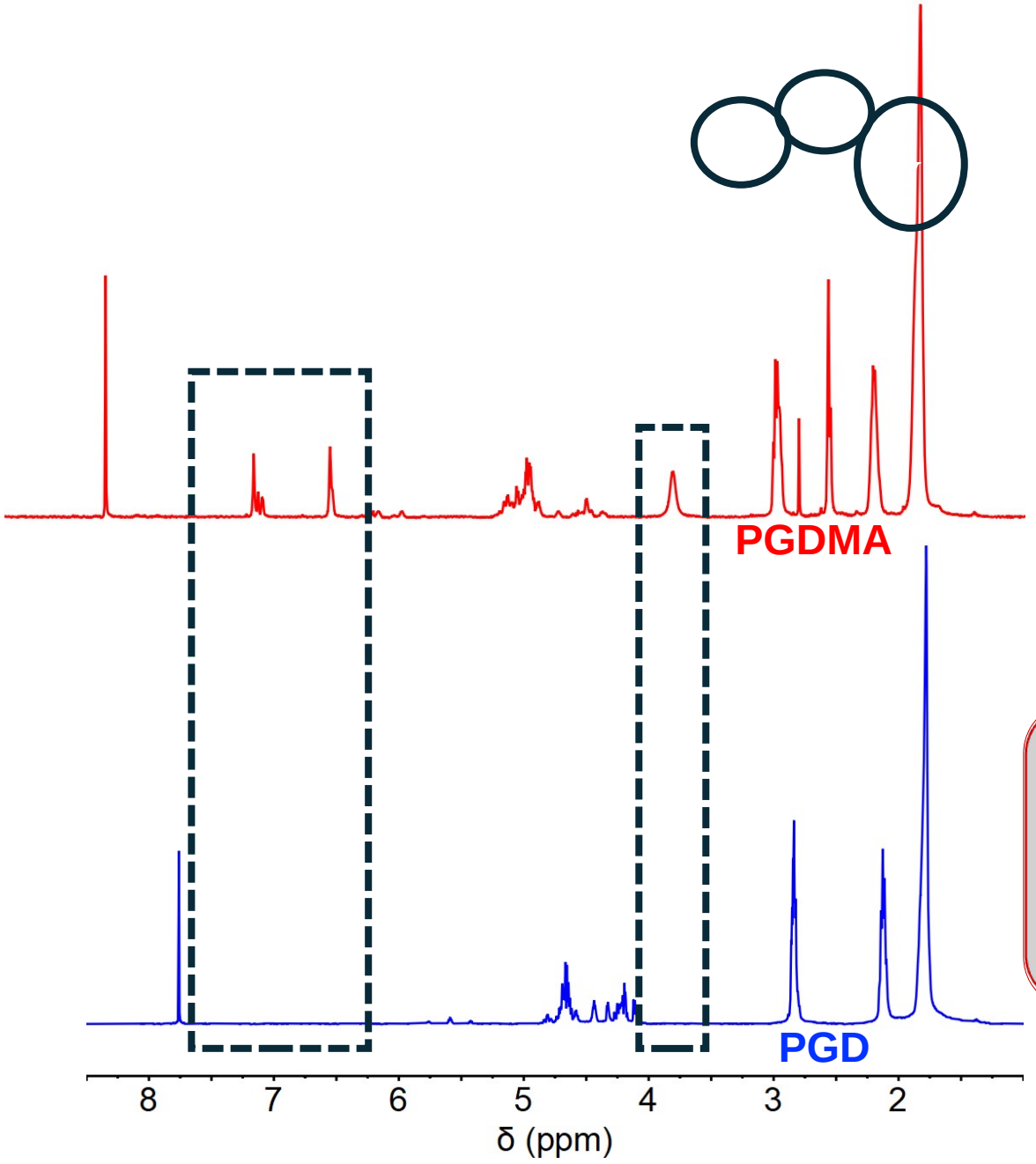
Our goal is to synthesize a shape memory polymer which is 3D printable, has T_g near body temperature and biocompatible.

Synthesis



Synthesis of PGDMA

NMR



Vinyl peaks from methacrylation= 6.18 ppm, 5.62 ppm

Methyl peaks from methacrylation=1.92 ppm

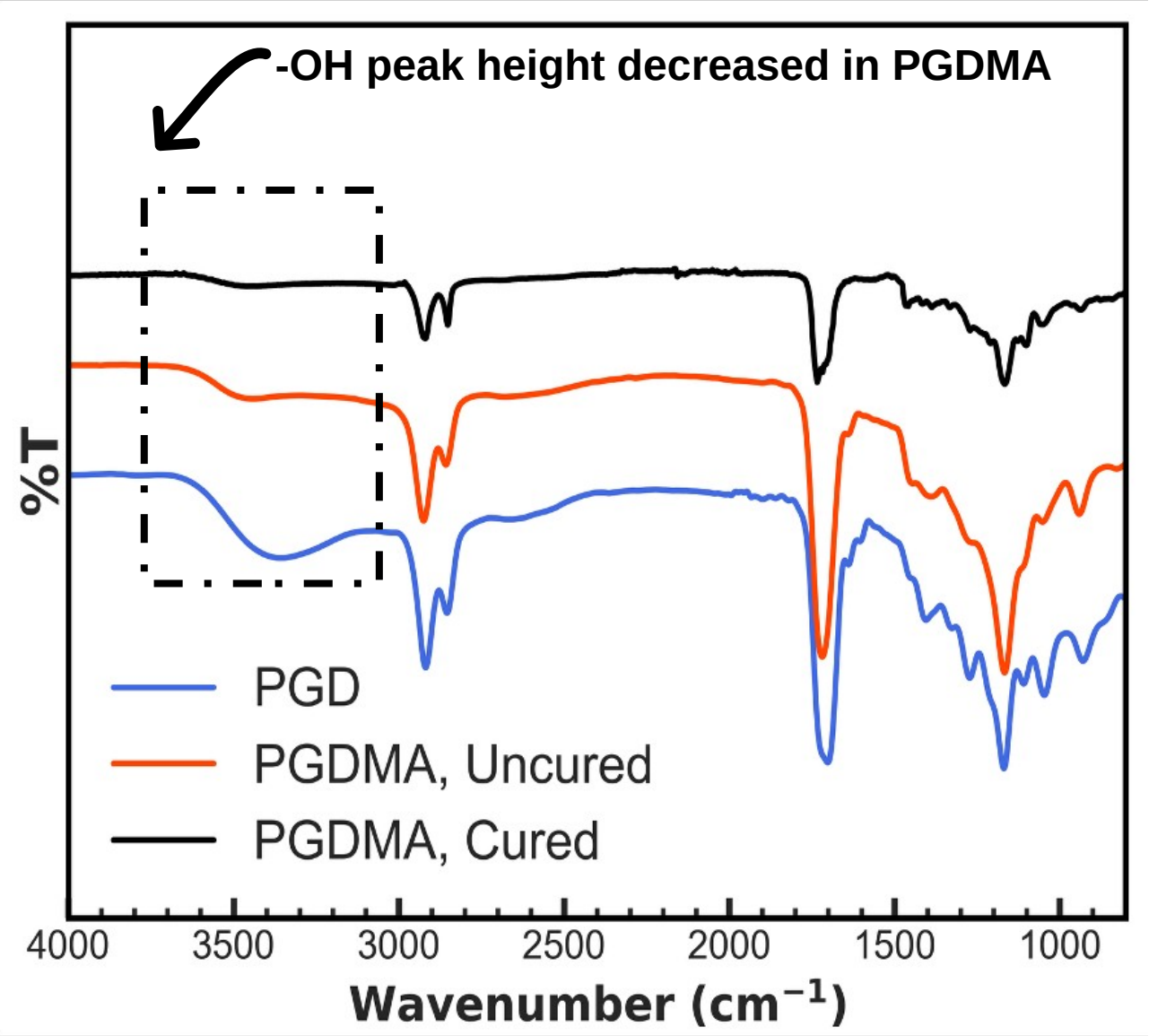
Dodecanedioic acid peaks for unique protons=1.62 ppm, 2.34 ppm &1.27 ppm

Degree of methacryaltion

$$:\frac{\text{Integral of vinyl peaks of methacrylation}}{\text{Integral of methylene peaks}}\times 100\%$$

=23.22%

FTIR

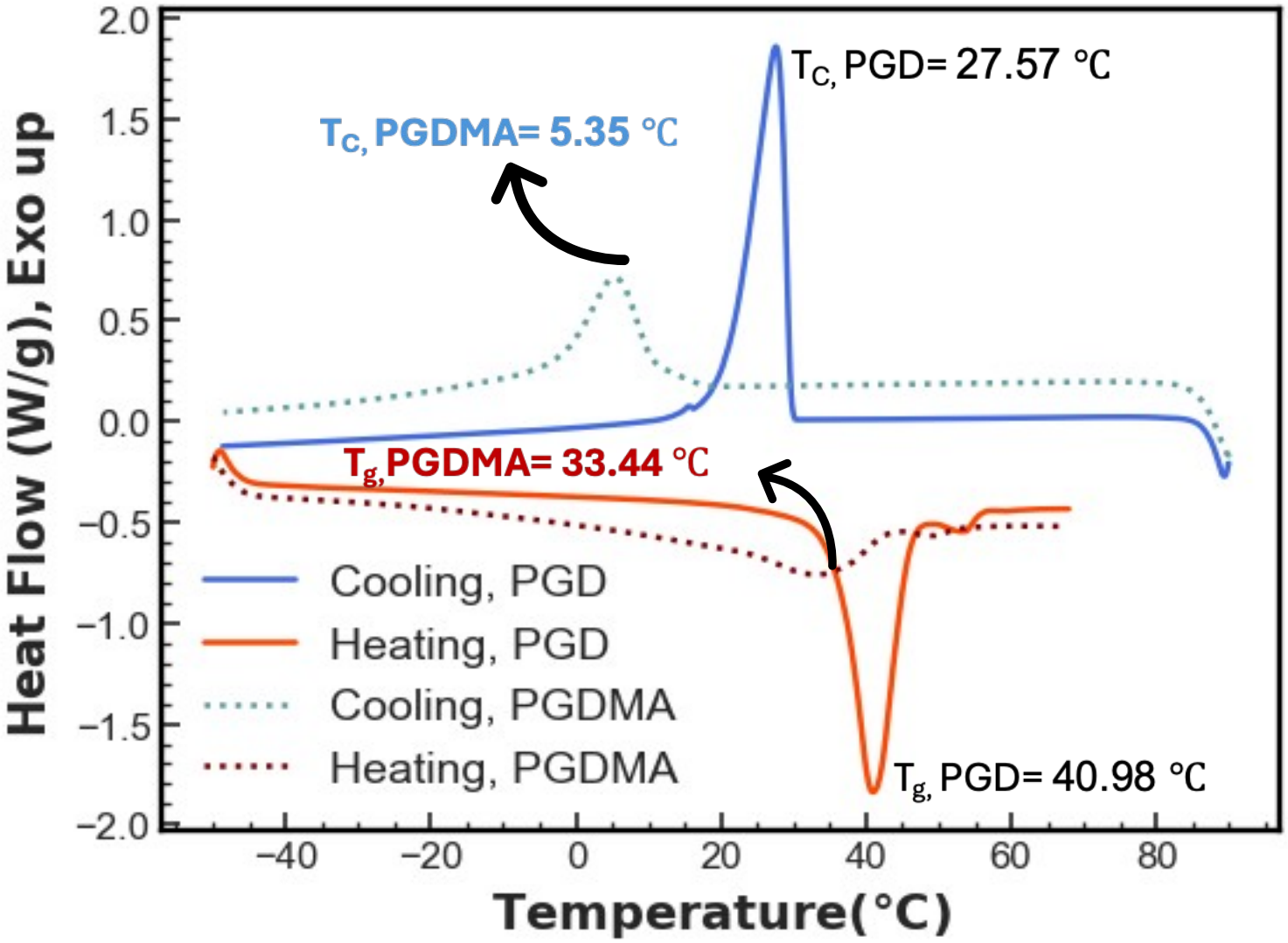


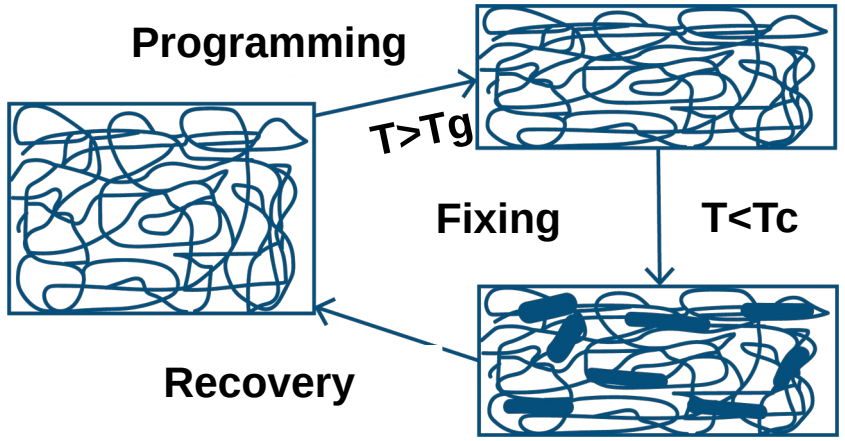
- Methacrylic Anhydride has replaced the secondary hydroxyl group in PGD

- –OH group intensity decreased in FTIR around 3300 cm^{-1}

Thermal Characterization

Will help us to determine the shape programming and fixing temperature

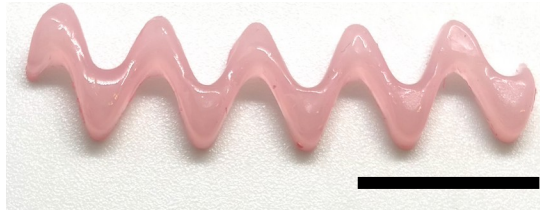




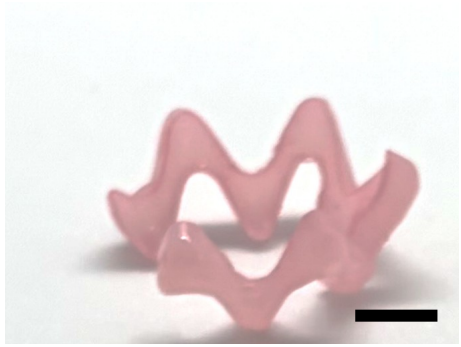
Shape Memory Visualization

Programming
(heating)

Fixing
(cooling)









Original Shape

Recovered Shape

Scale bar= 1 mm

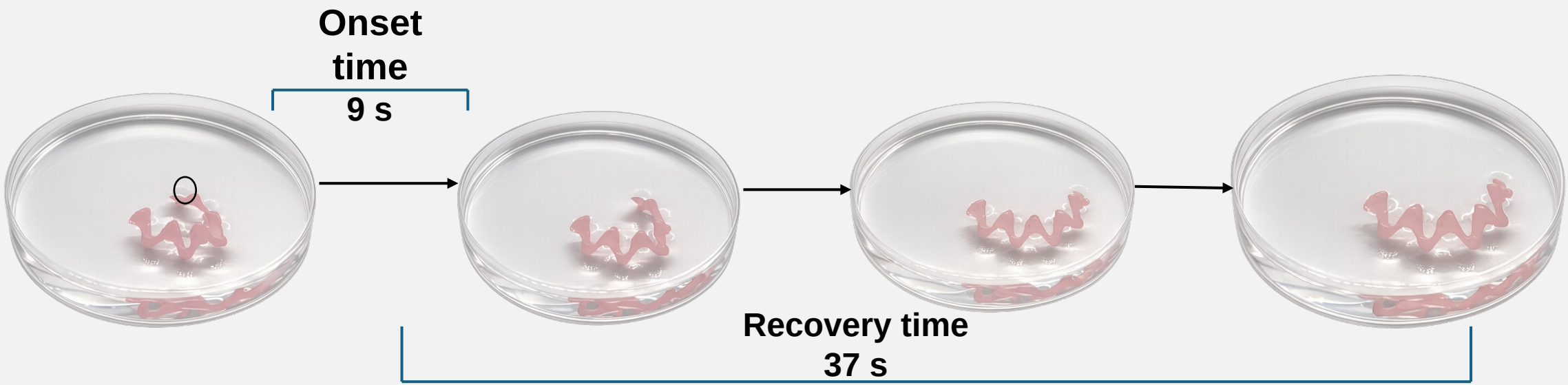
Programming Temperature= 80 ℃
Fixing Temperature= -20 ℃

Shape Memory Visualization

T=37 °C

1 cm

Onset and Recovery Time



Onset time can be programmed by varying the programming temperature.

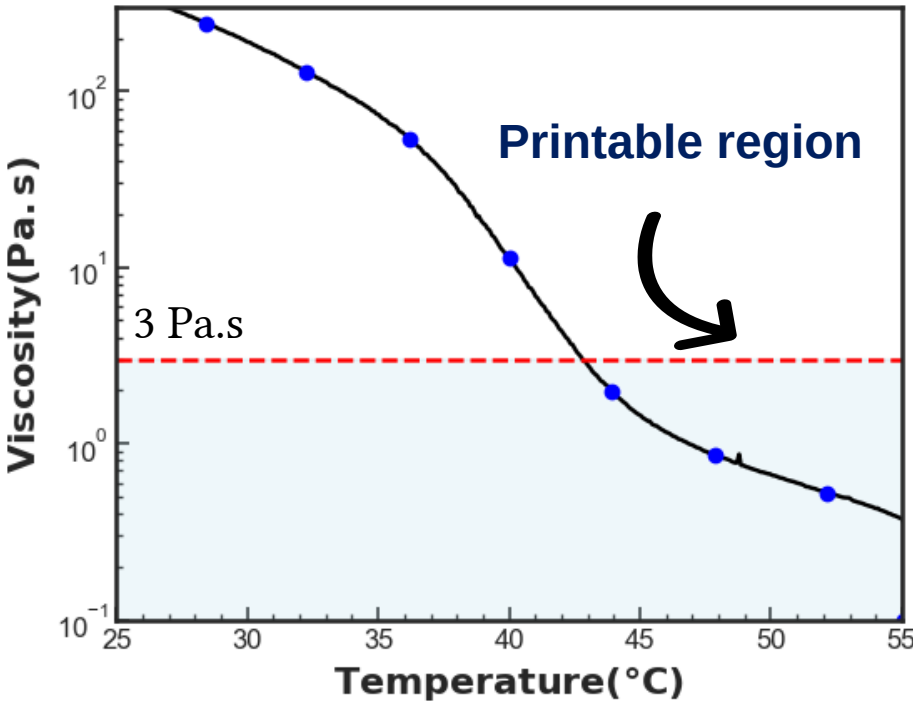
Rheology roadmap for evaluating a DLP ink

Rheology experiment

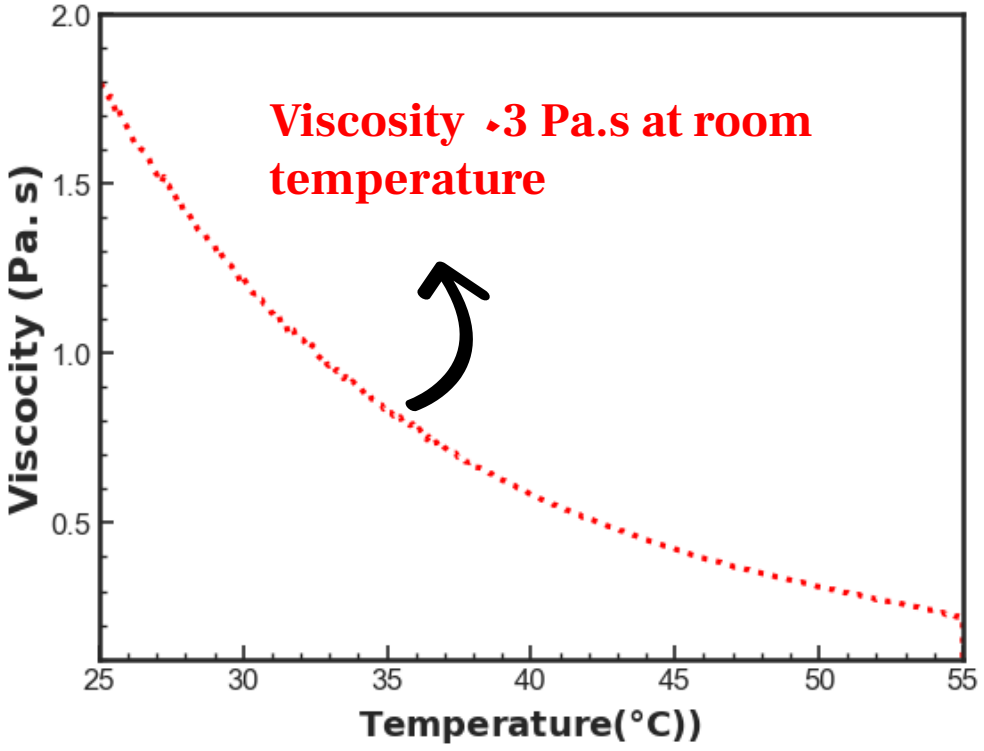
1.Viscosity < 3 Pa.s
2.UV crosslinking

Yes → Use as a DLP ink

No → Reformulate



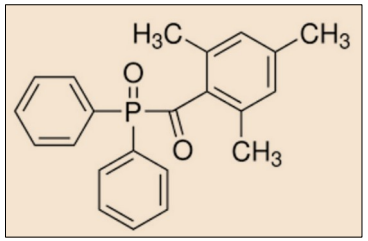
Reformulate



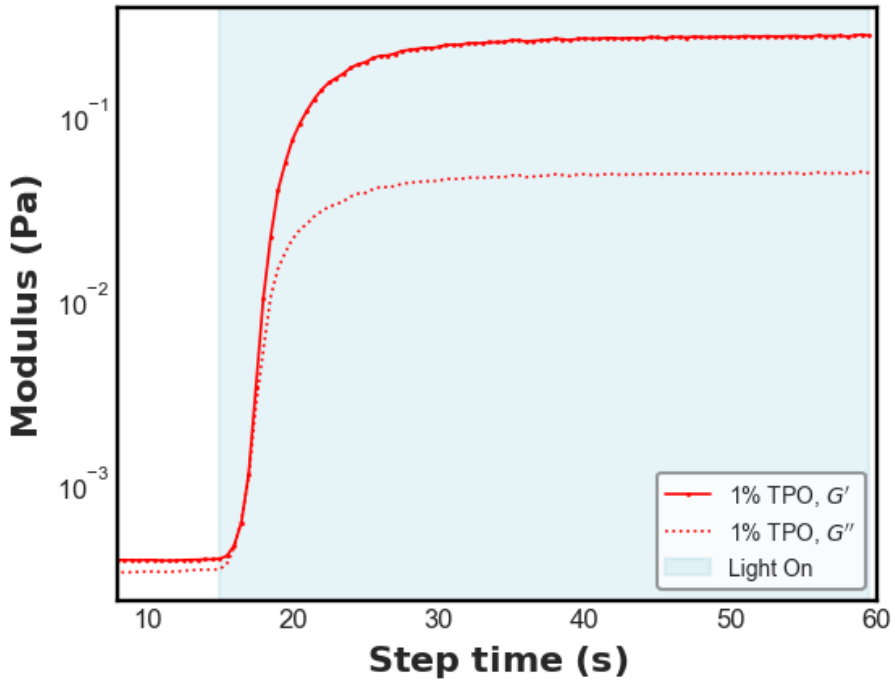
Photorheology



PGDMA resin

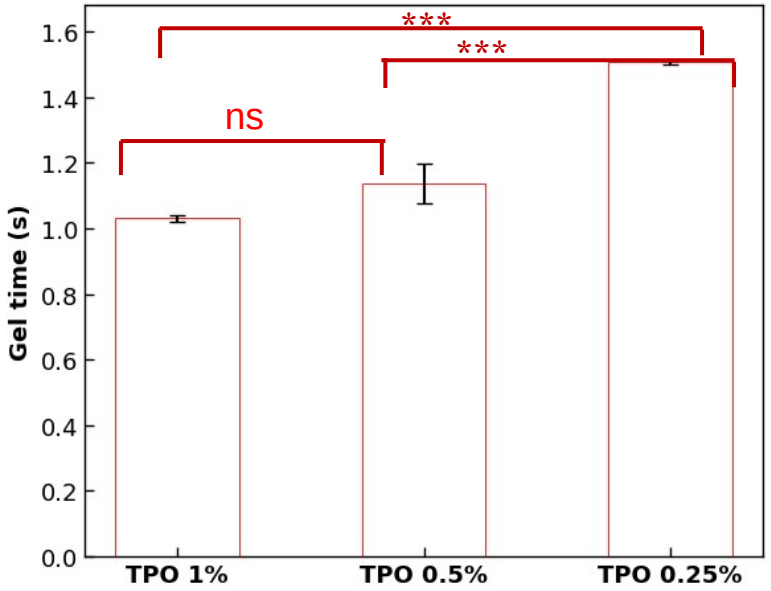


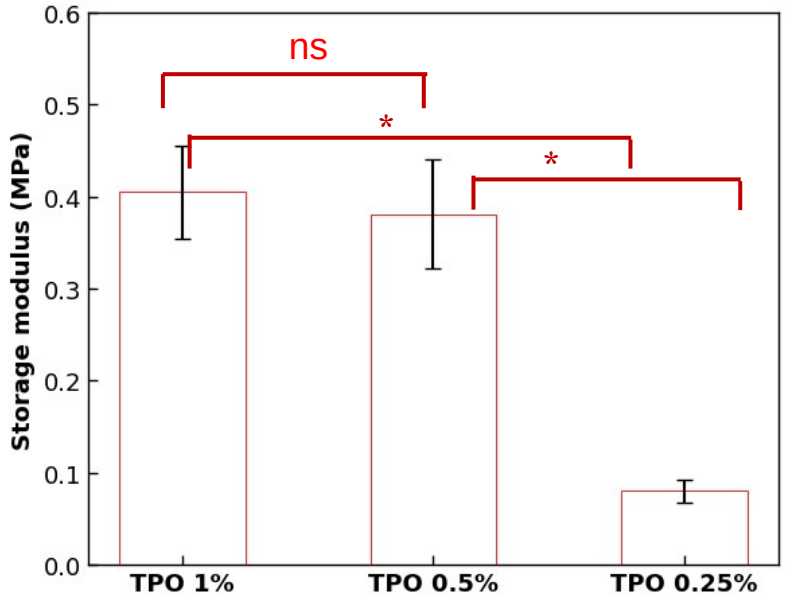
TPO



Rapid crosslinking while using 1% TPO as a photoinitiator.

PGDMA resin is 3D printable



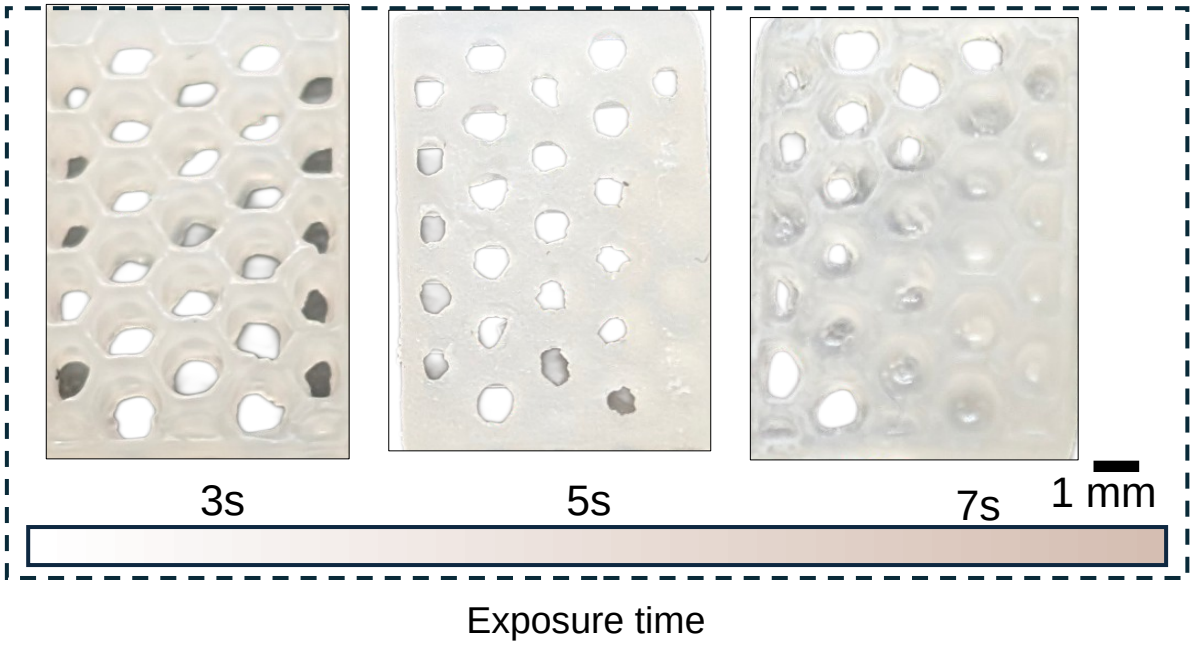


0.25% TPO creates softer network

Resin is fully crosslinked while using 0.5% TPO

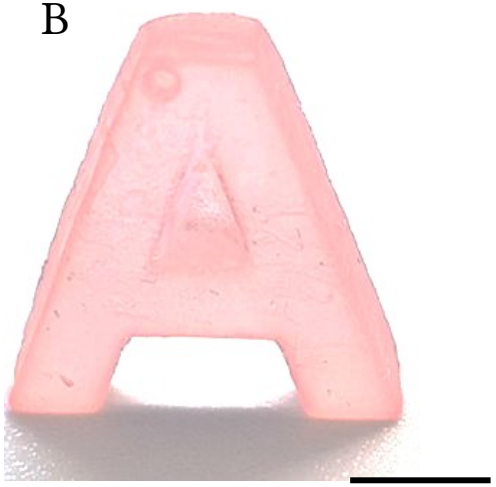
3D Printability

A

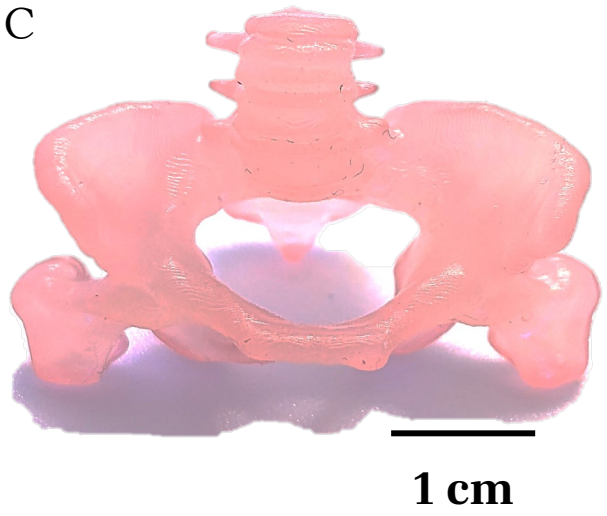


Resolution decreases with the exposure time

B



C

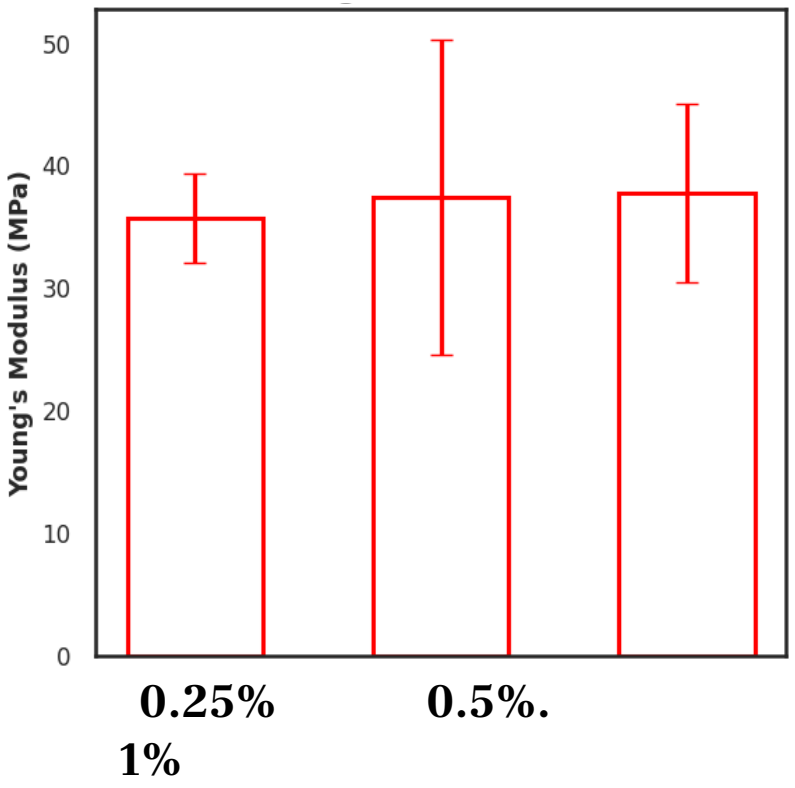


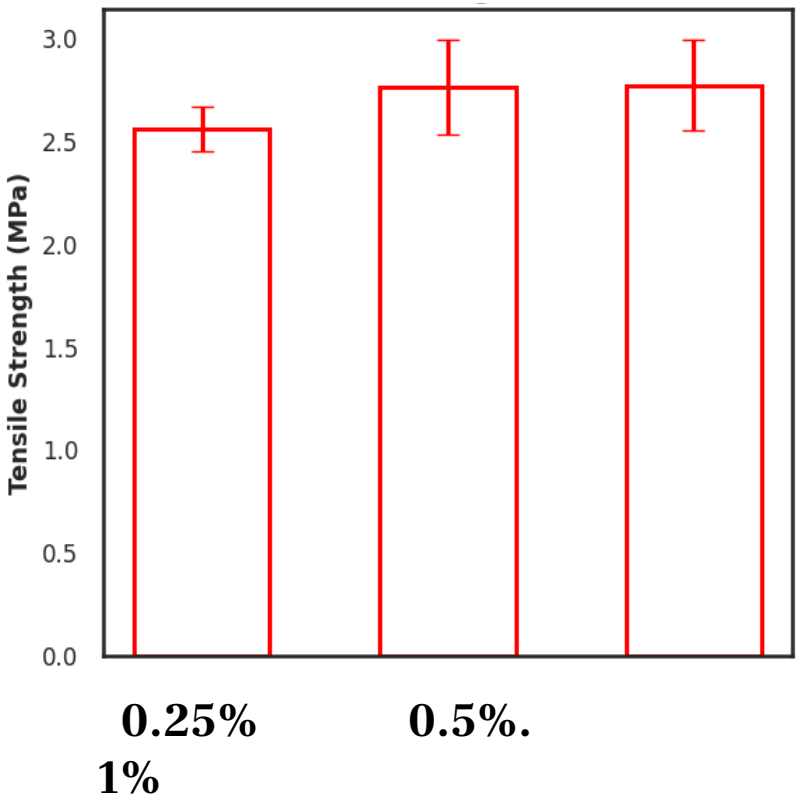
D

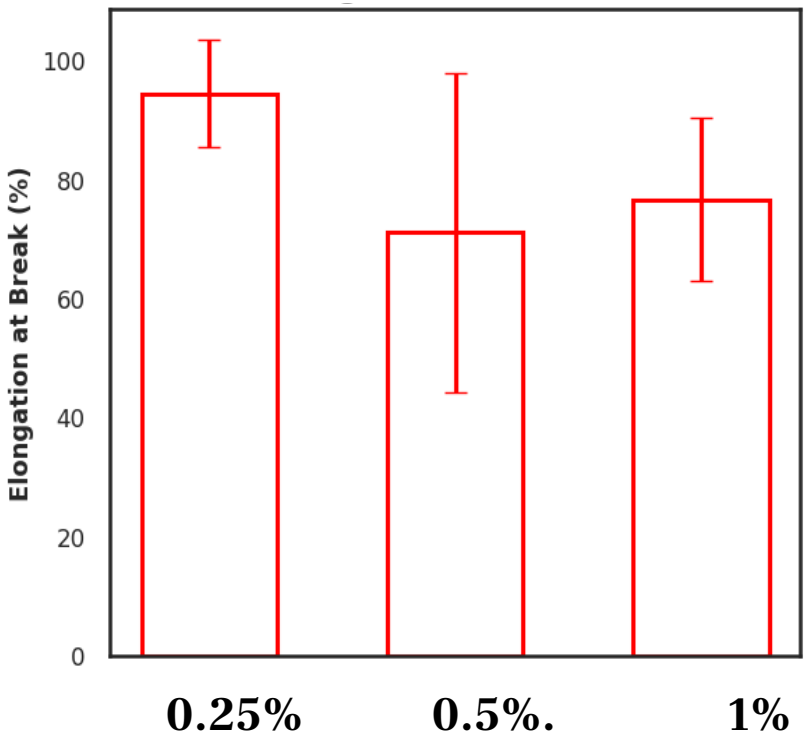


B. Letter A
C. Pelvic Bone
D. Flower Knot

Mechanical Characterization







Young's Modulus in PGDMA ~ 35 MPa. It can be used **for tendons.**

Elongation at break varies **0.78% to 97%** with decreasing %TPO.

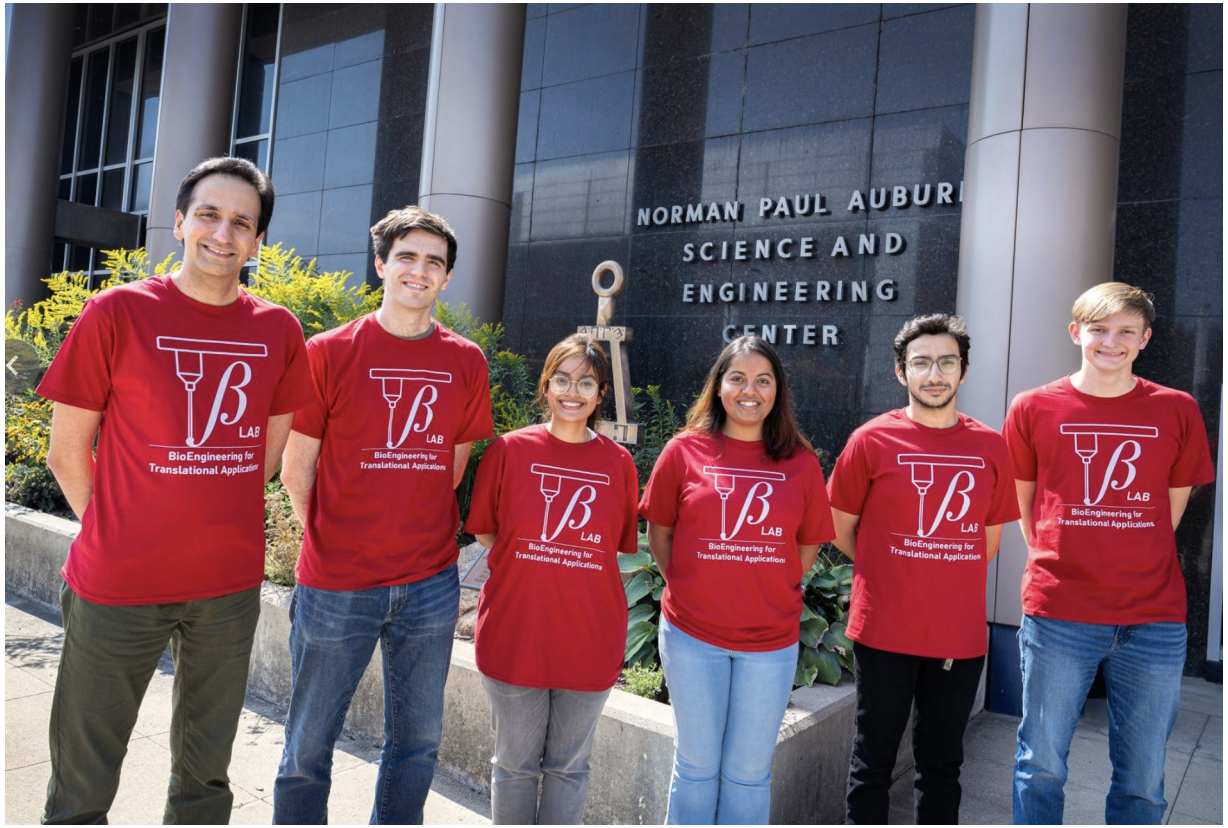
Discussion and Future works

- A novel polymer was developed and characterized with 3D printability and transition temperature near body temperature and shape memory properties
- Future work will include assessing the cytotoxicity and degradation properties of the polymer.

Thank You!

LAB

BioEngineering for Translational Applications



American Heart Association.

NSF National Science Foundation